

Predicting Hospital Readmission in Diabetic Patients Using Machine Learning: A Comparative Analysis

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Abstract— Hospital readmission among diabetic patients represents one of the most persistent and costly challenges in modern healthcare, carrying significant implications for patient wellbeing, clinical capacity, and care quality. This project applied logistic regression with LASSO regularization, random forest, and support vector machine classifiers to the problem of predicting 30-day readmission outcomes across three classes: no readmission (Class 0), readmission after 30 days (Class 1), and early readmission within 30 days (Class 2). Using 101,766 inpatient encounters drawn from the Strack et al. Health Facts dataset spanning 130 US hospitals from 1999 to 2008, each model was evaluated on overall accuracy and per-class recall, with particular attention to Class 2 as the most clinically urgent and most underrepresented outcome. Principal component analysis was additionally applied to examine the structure of the 98-dimensional feature space and assess whether the three readmission classes form geometrically distinct populations. Across all supervised configurations, overall accuracy was bounded between 0.49 and 0.58, with Class 2 recall consistently the hardest metric to improve regardless of algorithm, feature set, or balancing strategy. The RBF SVM achieved the highest Class 2 recall at 0.30, outperforming all linear models on minority class identification despite lower overall accuracy. PCA revealed that 56 components were required to explain 80% of total variance, and that the three readmission classes overlap almost completely in the two-dimensional projection, suggesting the prediction difficulty reflects the limits of structured discharge data rather than a failure of model selection.

I. INTRODUCTION

A. Background

Hospital readmission represents one of the most persistent and costly challenges in modern healthcare, placing strain on clinical inpatient capacity, institutional resources, and most importantly, on the patients cycling through repeated hospitalizations and outpatient interventions. The burden is not only physical and financial but also reflects a deeper failure in the continuum of care that patients depend on to maintain stability outside a clinical setting. Coming from a background in Health Psychology, I have always been drawn to questions at this intersection of human wellbeing and

systems of care, how illness is experienced, how care transitions succeed or break down, and what the data underlying those outcomes can tell us that individual clinical encounters cannot. During my literature search, I encountered the work of Strack et al., which clearly highlights the vulnerabilities of diabetic patients and those at risk for hyperglycemia. The readmission risk for this population is compounded by the complexity of glycemic control, the prevalence of chronic comorbidities, and the gaps that frequently emerge between initial discharge and follow-up care, making early readmission not just a clinical event, but a signal that something in the initial management of that patient's care was insufficient.

B. Literature Review

In "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," Strack et al. [1] performed a national assessment of inpatient diabetes care, extracting from a comprehensive clinical data warehouse representing ten years of care (1999 to 2008) across 130 hospitals throughout the United States. From an original database of over 74 million unique hospital encounters, the team identified approximately 70,000 inpatient diabetic encounters with sufficient clinical detail according to their inclusion criteria (inpatient admission, a recorded diabetes diagnosis, laboratory tests performed, and medications administered during the encounter). Using multivariable logistic regression, they modeled the relationship between HbA1c measurement and early readmission rates while controlling for covariates including patient demographics, type of admission, illness severity, and primary diagnosis. HbA1c, or glycated hemoglobin, reflects average glycemic control over the preceding two to three months and serves as a standard marker of diabetes management quality, making the act of test ordering a reasonable proxy for the degree of clinical attention given to a patient's diabetes during hospitalization. Their analysis yielded three key findings. First, HbA1c was ordered infrequently, occurring in only 18.4% of eligible encounters.

Second, when the test was not ordered, providers still changed diabetes medications in 42.5% of cases; when the test was ordered, medication change rates rose to 55%, suggesting that ordering the test prompted more active clinical management. Third, and most relevant to the present project, simply measuring HbA1c was associated with a lower readmission rate, but only for patients whose primary diagnosis was diabetes. For patients admitted primarily for circulatory or respiratory conditions who also had diabetes, this association did not hold, suggesting that the protective effect depended on diabetes being the primary clinical focus of the encounter.

C. Project Objectives

While Strack et al. examined a specific clinical hypothesis about one biomarker using multivariable logistic regression, this project takes a broader and more updated approach to the same problem. Their paper was published in 2014 using hospital data from 1999 to 2008, and their analysis was designed to answer one focused statistical question about HbA1c. My goal was different. I wanted to take this same dataset and ask a more practical question: given everything we know about a patient at the time of their discharge, can we actually predict whether they will come back? And if so, how soon? To answer that, I applied three machine learning classification models, Logistic Regression with LASSO Regularization, Random Forest, and Support Vector Machine, each trained to predict one of three outcomes for every patient encounter: no readmission (Class 0), readmission after 30 days (Class 1), or readmission within 30 days (Class 2). The early readmission group, Class 2, is the most clinically urgent outcome and the hardest to predict given how underrepresented it is in the data. Comparing how each model handled that challenge was a central part of the analysis. Beyond the supervised models, I also applied Principal Component Analysis to step back and ask a more fundamental question about the data itself. With 98 features present after preprocessing, I wanted to understand how much of the variation across patients is actually captured by a small number of underlying patterns, and whether patients in different readmission classes look meaningfully different from each other in that compressed space.

II. DATASET

A. Source and Collection

The dataset used in this project is the extracted and curated version produced by Strack et al. [1] from the Health Facts database (Cerner Corporation), now publicly available through the University of California, Irvine (UCI) Machine Learning Repository. As previously discussed, it represents ten years of inpatient care (1999 to 2008) across 130 hospitals spanning all four U.S. census regions (58 in the Northeast, 18 in the Midwest, 28 in the South, and 16 in the West). After applying the five inclusion criteria established by Strack et al., 101,766 encounters were retained for analysis. It should be noted that Strack et al.'s final analytical dataset comprised 69,984 encounters after additionally excluding patients who died during the encounter or were discharged to hospice, as these populations physically cannot be readmitted and their inclusion would artificially suppress observed readmission

rates. In the present project, these exclusions were handled during the preprocessing stage, which accounts for the difference in total encounter count. Prior to preprocessing, an independent verification script was developed to confirm all descriptive statistics cited in this section directly against the raw dataset.

B. Feature Breakdown

The dataset contains 47 original features organized into four categories: patient demographics, admission and discharge characteristics, clinical encounter variables, and medication history. Patient demographics include race, gender, and age. Race spans five categories (Caucasian, African American, Hispanic, Asian, and Other), with Caucasian patients making up 74.7% of the dataset. Gender is recorded as male, female, or unknown/invalid. Rather than recording exact age, patients are grouped into ten-year intervals from [0-10) through [90-100). The dataset skews older, with a median age of 64.9 years and 66.2% of patients over the age of 60, which reflects the higher prevalence of Type 2 diabetes in older populations.

Admission and discharge characteristics describe how and where a patient entered and left the hospital. Three of the four features in this group (admission type, discharge disposition, and admission source) are stored as numeric codes in the raw data, which I decoded using the Integer Identifier System (IDS) mapping file provided with the dataset. Admission type tells us the urgency of the admission, such as Emergency, Urgent, or Elective, with Emergency being the most common at 53.3% of encounters. Discharge disposition describes what happened when the patient left, ranging from going home (63.4% of encounters) to being transferred to another facility. Admission source describes how the patient arrived, whether through a physician referral, the emergency room, or a transfer. Time in hospital is the only numeric variable in this group, ranging from 1 to 14 days with an average stay of 4.3 days.

Clinical encounter variables capture what actually happened during each hospitalization. Six of the eleven features are numeric counts: how many lab tests were run, how many procedures were performed, how many medications were given, and how many times the patient had visited the hospital in the prior year across outpatient, emergency, and inpatient settings. These prior visit counts are especially useful for predicting readmission because they give a sense of how much healthcare a patient was already using before this admission. Two features record lab results in categories rather than exact values, specifically the glucose serum test result and the HbA1c result. The remaining three features are the patient's primary, secondary, and additional secondary diagnoses coded using the International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9-CM) System. It is worth noting that even though every patient in this dataset has diabetes, it is not always the main reason for their admission. Someone admitted for a heart attack who also has diabetes would have a circulatory disease code listed as their primary diagnosis, with diabetes appearing as a secondary condition. Because each diagnosis field contained over 700 unique codes, I grouped them into broader disease categories during preprocessing, which is described in

Section III. A count of total diagnoses per encounter is also included as a general indicator of how medically complex a patient's situation was.

Medication history is covered by 26 features. Twenty-four of these track individual diabetes medications prescribed during the encounter, including common drugs like metformin, insulin, glipizide, and glyburide, as well as combination medications. Each is recorded as one of four values: No (not prescribed), Steady (prescribed, no change), Up (dose increased), or Down (dose decreased). This tells us not just whether a medication was given, but whether the care team was actively adjusting it. The remaining two features give a high-level summary: change flags whether any diabetes medication was adjusted during the stay, and diabetesMed flags whether any diabetes medication was prescribed at all.

C. Target Variable and Class Distribution

The target variable is readmitted, which takes one of three values: no readmission (Class 0), readmission after more than 30 days (Class 1, >30), and readmission within 30 days (Class 2, <30). The 30-day cutoff is a widely used healthcare standard for measuring whether a discharge was handled well. The class breakdown confirmed directly from the raw data showed approximately 54% of encounters resulted in no readmission, 35% in readmission after 30 days, and only 11% in early readmission within 30 days. That 11% was the group that mattered most clinically, because it suggests the patient was sent home before they were truly stable. It was also the hardest group to predict because it was so underrepresented in the data. A model that simply guessed no readmission every time would still have been right 54% of the time, but it would never have caught a single early readmission case, making it useless in practice. Getting the models to meaningfully identify that 11% rather than just chasing overall accuracy was one of the central challenges of this project. Before any modeling could begin, the raw data also required significant cleaning, including handling missing values, simplifying the diagnosis codes, encoding categorical variables, and scaling the features, all of which are covered in the next section.

III. DATA PREPROCESSING AND EXPLORATION

A. Exploratory Data Analysis

Prior to preprocessing, I performed exploratory data analysis on the raw dataset to understand the distributions of key variables, identify potential challenges, and inform succeeding cleaning decisions. All summary statistics gathered in this section were computed directly from the raw dataset before any transformations were applied.

B. Patient Demographics

Fig. 1. Patient demographics across race, gender, and age group distribution.

Here I break down the patient population across race, gender, and age. Looking at race, the dataset is heavily skewed toward Caucasian patients at 74.8% of encounters, with African American patients making up 18.9%. Every other group (Hispanic, Other, and Unknown) falls below 3% individually, which is worth keeping in mind when interpreting any race-related findings. Gender was balanced, with female patients

at 53.8% and male at 46.2%. The Unknown/Invalid category was essentially negligible at 0.0%. The age distribution was one of the more telling findings in this section. Encounter volume climbs steadily from the youngest age groups, peaks in the 70-80 interval, and drops off sharply after that. 66.2% of patients were over 60, which lines up with what was observed in the original article about the demographics of Type 2 diabetes and is something I kept in mind when thinking about how well these models could generalize beyond this specific population.

C. Readmission Class Distribution

Fig. 2. Readmission class distribution showing class counts and proportions.

The breakdown of the target variable across all three readmission classes revealed a significant class imbalance that was one of the first challenges I had to address. Of the 101,766 encounters, 53.9% resulted in no readmission (Class 0), 34.9% in readmission after 30 days (Class 1), and only 11.2% in early readmission within 30 days (Class 2). This matters because a model that simply predicted no readmission for every patient would still be right about 54% of the time, while never once correctly identifying an early readmission case. That kind of model would be completely useless in a clinical setting. The early readmission class is not just the smallest but also the most clinically important, since a patient readmitted within 30 days was likely discharged before they were truly stable. This imbalance directly shaped every modeling decision I made in the sections that follow, particularly around class weighting.

D. Numeric Variable Distributions

Fig. 3. Distribution of numeric variables in the diabetic dataset using log-scale x-axis.

Looking at the distributions of the eight numeric variables revealed a consistent pattern of right skew across the utilization-related features. I applied a log scale to the x-axis throughout because outpatient visits, emergency visits, and inpatient visits in the prior year were all extremely right skewed. Most patients had zero or one prior visit in each category, but a small number had many, creating a long right tail that makes the distribution nearly unreadable at a linear scale. The log scale compresses that tail enough to see the shape. Number of lab procedures spreads widely across many values, which tells me that clinical complexity varied quite a bit across encounters. Number of medications looks roughly bell-shaped and centered around 10 to 15 medications, suggesting a fairly consistent level of medication management across most hospitalizations. One thing that stood out in number of diagnoses was the sharp spike at 9, which reflects a data entry ceiling in the original system rather than anything clinically meaningful, and I noted this as a data artifact. Time in hospital sits mostly in the lower range, which makes sense given the 1-to-14-day inclusion criterion. Overall, the heavy skew in the utilization variables confirmed that StandardScaler was the right choice during preprocessing, since it standardizes everything to a common scale regardless of the original distribution.

E. Data Cleaning and Preprocessing

The first thing I did was replace all instances of ‘?’ with ‘NaN’, since that is how this dataset encoded missing values rather than leaving cells blank. From there I looped through every column to count how many values were missing and what percentage of the column that represented. For weight (96.86% missing), payer code (39.56%), and medical specialty (49.08%), I dropped the columns entirely. Weight was too sparse to be useful and had no clear relationship to readmission risk. Payer code had no clinical significance. Medical specialty was the more interesting case. Unlike Strack, who kept it and imputed “missing” as a category itself, I dropped it because a blank specialty field does not actually tell us anything about how a patient was managed. All three features fall under Missing Completely at Random (MCAR), meaning their absence carries no information, so I decided removing them was the best decision. For race, max glucose serum, and HbA1c, I used special value imputation rather than a statistical method. Race was filled with “Not recorded” because an undocumented race is a documentation gap, not a truly absent data point. Max glucose serum and HbA1c were filled with “Not tested” because these tests were not missing in the traditional sense as they simply were not ordered. Filling them with “Not tested” preserves that clinical meaning rather than hiding it. For the three diagnosis fields (diag_1, diag_2, diag_3), which had less than 2% missing in each, I used mode imputation. The values were sparse enough that anything more complex like K-Nearest Neighbors (KNN) would have added unnecessary work for minimal benefit.

F. Transformations

Age was stored as string intervals in the raw data. Because age has a real ordinal structure, I mapped the ten intervals to integers 0 through 9 rather than one-hot encoding them, as that would have treated each age group as a completely separate and unrelated category, discarding the ordering that actually matters here.

The three diagnosis fields each had over 700 unique ICD-9-CM codes, which is too many categories to model directly. I wrote a grouping function that collapsed everything into eight clinically meaningful disease categories: Diabetes, Neoplasms, Circulatory, Respiratory, Digestive, Genitourinary, Musculoskeletal, and Injury. Anything that did not fit went to Other. This includes E and V codes, which are supplementary rather than primary diagnoses. E codes describe how an injury happened, not the injury itself, and V codes describe healthcare contacts for non-illness reasons like a routine checkup or a medication refill.

For the target variable, I used label encoding to map the three outcomes to integers: NO to 0 (Class 0, no readmission), greater than 30 days to 1 (Class 1), and less than 30 days to 2 (Class 2, early readmission). One thing worth noting is that mapping these outcomes to the numbers 0, 1, and 2 technically implies an order that does not really exist here. Being readmitted within 30 days is not “more” than being readmitted after 30 days in any meaningful way. They are just different outcomes. However, this is a standard approach for multiclass problems, and the models do not actually treat these numbers as a scale.

Gender was binary encoded directly, with Male mapped to 1 and Female to 0. All remaining categorical variables were one-hot encoded with one redundant category removed per feature to prevent perfect multicollinearity. The resulting boolean columns were converted to integers for compatibility with the modeling pipeline. This encoding process is what expanded the feature count from the original 47 columns to 98. One-hot encoding works by creating a new binary column for each unique category in a variable, so a feature like race with five categories becomes four new columns. Across all categorical variables in the dataset, this expansion drove the feature count from 47 to 98, and it is the 98-column matrix that all three models and PCA were trained and evaluated on.

G. Train/Test Split and Feature Scaling

Once everything was cleaned and encoded, I separated the feature matrix X from the target variable y and dropped the encounter ID and patient number columns since they are administrative identifiers with no predictive value. The dataset was divided into a 70/30 train/test split, reserving 70% of the data for training the models and 30% for evaluating their performance on unseen data.

Scaling was done using StandardScaler, which transforms each feature to have a mean of zero and a standard deviation of one. The key detail here is that I fit the scaler only on the training data and then applied that same transformation to the test set. If I had fit it on the full dataset before splitting, the scaler would have learned statistics from the test set and used them during training, which would result in data leakage. That would have made the model look better than it actually is on data it had never seen. The cleaned and scaled data was saved and passed forward to each modeling script that followed.

IV. MODEL DEVELOPMENT

A. Logistic Regression and LASSO Feature Selection

The first model I built was Logistic regression (LR), starting with a baseline before introducing any feature selection via Least Absolute Shrinkage and Selection Operator (L1 regularization, LASSO). Logistic regression essentially tries to find the best linear boundary (or in the case of higher dimensions, a flat boundary) that separates my target classes from each other. Rather than predicting a raw number, it outputs a probability for each class and assigns the patient to whichever class scores highest. Because the dataset is so imbalanced, I used `class_weight='balanced'`, which essentially tells the model to care more about getting the minority class right during training, even if that hurts overall accuracy. My baseline model output 49% overall accuracy. To evaluate per-class performance I relied on two key metrics from the classification report. Recall measures the proportion of actual positives that the model correctly identified, so a Class 2 recall of 0.39 means the model correctly flagged 39% of true early readmission patients. Precision measures the proportion of predicted positives that were actually correct, so a precision of 0.19 for Class 2 means that when the model did flag an early readmission, it was right 19% of the time. The F1 score is the harmonic mean of precision and recall,

combining both into a single number that penalizes imbalanced tradeoffs between the two. A Class 2 recall of 0.39 is meaningful for a population that only makes up 11% of the total data. Without the balanced weighting, the model just learns to predict no readmission almost every time.

With the baseline established, I moved onto feature selection. Not all 98 features contribute equally to prediction and including redundant or low-signal features can add noise without improving performance. This is where LASSO (L1) comes in. L1 is a form of regularization that adds a penalty to the model during training that pushes less useful feature weights all the way to zero, effectively cutting them out. The strength of this penalty is controlled by parameter C , where a smaller value means more aggressive cutting. I tested three values (0.1, 0.01, 0.001) to see how far I could reduce the feature set before performance started to suffer. At $C=0.1$ the model kept all 98 features. At $C=0.01$ it dropped 15 and kept 83 features, with an accuracy of 0.4884. At $C=0.001$ it dropped almost everything, only retaining 9 features, with accuracy changing by just 0.0001 (0.4883). That minimal drop across such a dramatic feature reduction shows that a lot of the features created during preprocessing were either redundant or just not carrying useful information for this analysis. I went with $C=0.01$ as my final choice, keeping 83 features. Cross-validation did recommend $C=0.001$, but I decided to push back on that. Getting 9 features instead of 83 for essentially no accuracy gain means the model was most certainly cutting features that have real clinical meaning. $C=0.01$ keeps a richer, more clinically grounded feature set which felt like the right call overall.

B. Random Forest

Building on decision trees, Random Forest (RF) utilizes many decision trees, each trained on a slightly different random sample of the data (bootstrap dataset) and using a random subset of features at each node split. None of the individual trees are that useful on their own, but by averaging out their predictions through the majority vote model, the errors tend to cancel out, and the combined results should be much more stable. It can also pick up on non-linear patterns and interactions between features that logistic regression cannot find due to straight-line boundary limitations. Establishing a baseline with 100 trees and no class weighting achieved 58.1% overall accuracy. Class 0 (no readmission) had an F1 of 0.71 and Class 1 (readmission after 30 days) came in at 0.42. Class 2 (early readmission) recall was essentially zero at 0.01, meaning the model almost never correctly identified an early readmission patient despite a precision of 0.46. This is the same pattern seen in every unbalanced baseline across all three models. The model learned that predicting no readmission most of the time was the easiest path to high accuracy, completely at the expense of the class that matters most clinically.

With a baseline established, I introduced class balancing and stepped through 50, 100, and 200 trees on both the full feature set and the LASSO-selected 83 features. Across all six configurations, accuracy stayed tightly clustered between 0.57 and 0.58, with the best result coming from 200 trees on the full feature set at 0.5825. Class 2 recall

remained at 0.01 across every configuration regardless of tree count or feature set. That flat performance told me that adding more trees was not the bottleneck. The forest stabilized quickly and the ceiling was determined by the data, not the model complexity. The LASSO feature set performed nearly identically to the full feature set at every tree count, which is consistent with the LR results and confirms that the 15 dropped features were genuinely not contributing useful signal. The reason class weighting helped so much more in LR than RF comes down to how each model applies it. In LR, the balanced weight rescales the loss function globally so every single Class 2 error costs more everywhere during training. In RF, each tree is only trained on its own bootstrap sample, and since Class 2 patients are only 11% of the data to begin with, most bootstrap draws still contain very few of them regardless of how those rows are weighted. The balancing helps at the margin but cannot overcome the structural underrepresentation built into the sampling process itself.

One significant advantage of Random Forest over the other two models is its built-in feature importance output, which measures how much each variable contributed to reducing prediction error across all 200 trees. The feature importance plot (Fig. 4) shows the mean decrease in impurity per variable.

Fig. 4. Random Forest top 15 feature importances by mean decrease in impurity.

The top features were `num_lab_procedures` (0.097), `num_medications` (0.086), `time_in_hospital` (0.063), `age` (0.055), `number_diagnoses` (0.044), `num_procedures` (0.044), and `number_inpatient` (0.043). Looking at that list, it reads like a profile of how sick a patient was during their admission. More lab tests, more medications, longer stays, more diagnoses, more prior inpatient visits. These are all signals of a more complex and acutely ill patient, and it makes complete clinical sense that illness severity would be the strongest driver of whether someone comes back. The appearance of `discharge_disposition_id` (0.043) in the top eight is also notable. It suggests that where a patient goes after leaving the hospital carries its own predictive weight independently of how sick they were, which connects directly to the care transition argument made in the introduction. Importantly, the top features identified here closely mirror what LASSO retained at $C=0.01$, providing convergent evidence from two completely different selection methods that these variables genuinely matter.

C. Support Vector Machine

SVM approaches classification differently from the prior two models. Rather than trying to classify every training point correctly, it focuses specifically on finding the widest possible gap between the classes, defined by the data points that sit closest to the boundary on either side. Those boundary points are called support vectors, and they are the only ones that determine where the line gets drawn. The parameter C controls how strictly the model enforces correct classification during training. A large C forces the boundary to hug the training data closely, which risks overfitting. A smaller C allows some mistakes in exchange for a wider, more robust

boundary. The shape of that boundary is determined by the kernel, which is one of SVM's most useful and flexible features.

A linear kernel finds the best straight line (or flat plane in higher dimensions) through the feature space, which works well when the classes are roughly linearly separable. An RBF kernel, which stands for Radial Basis Function, mathematically transforms the data into a higher-dimensional space where a linear boundary can capture curved, non-linear patterns that would be impossible to separate with a straight line in the original space. In practice this means RBF can model more complex relationships between features, which is why it is the most commonly used kernel for real-world tabular data. For the linear kernel I used LinearSVC, which is designed for large datasets and runs significantly faster. For the RBF kernel, I ran into a practical problem. RBF SVM gets exponentially slower as the dataset grows and running it on the full 71,236 training rows was not feasible. Instead, I trained the RBF model on a random 10,000-row subsample, while always evaluating on the full test set, which is noted as a methodological caveat when interpreting results.

With the setup in place, I ran the baseline LinearSVC with no class weighting first, which achieved 57.1% accuracy but completely missed Class 2 (early readmission) with zero recall, consistent with every other unbalanced baseline in this project. When I introduced class balancing at $C=0.1$, overall accuracy dropped to 56.0% but Class 2 recall came up to 0.12 and F1 to 0.16, a meaningful improvement over the zero-recall baseline even if the absolute numbers remain modest. Switching to the RBF kernel on the subsample gave the most surprising result of any model in this project. Overall accuracy dropped to 49.0%, worse than both LR and linear SVM, but Class 2 recall hit 0.30, the highest of any configuration I tested. The non-linear boundary that RBF draws is apparently better at capturing the information that distinguishes early readmission patients, even at the cost of more overall misclassifications. In a clinical setting where the primary goal is to flag patients most at risk of coming back within 30 days, that higher recall may actually be the more meaningful result.

D. Principal Component Analysis

It is worth addressing a methodological note about where PCA appears in this project. In a conventional machine learning pipeline, PCA is typically applied before the supervised models. That is, if its purpose were to reduce the feature space and feed a compressed representation into training. I placed it after the supervised models because that was not how I used it. Rather than using PCA-transformed features as input to Logistic Regression, Random Forest, or SVM, I applied it as a diagnostic tool after the fact to understand why all three models converged on a similar accuracy ceiling regardless of algorithm or configuration. In that framing, presenting PCA after the supervised results felt more honest: here are the results, and here is what PCA revealed about why they looked the way they did. I also want to be transparent that this was my first time working with PCA. I had covered it conceptually in class but had not applied it to a real dataset before this project. Most of what I

learned about implementing it, interpreting the scree plot, and reading the feature loadings came from working through it directly here, consulting the scikit-learn documentation, revisiting my lecture notes, and iterating until the output matched what I understood the method to be doing.

PCA was the last and only unsupervised method used in this project. After one-hot encoding, the dataset had 98 features to describe each patient encounter. The question PCA was applied to answer is whether all 98 of those features are actually necessary, or whether a shorter, cleaner representation can capture most of the same information. It does so by finding the directions in the feature space where patients vary the most from each other and creating a new set of axes (principal components), all of which are ranked by how much variation each one captures. The first component points in the direction of maximum spread across all patients, the second captures the most remaining spread after that, and so on. It is worth distinguishing PCA from L1 regularization here because while both reduce dimensionality, they do so in fundamentally different ways. LASSO keeps the original features and cuts the ones that hold no predictive weight. PCA combines all of them into a completely new set of axes designed to capture as much of that variation as possible in as few dimensions as possible. The result is not a subset of the original features but a transformed representation of it. Additionally, because PCA is unsupervised it has no knowledge of readmission outcomes at all. It is purely looking at how patients vary from each other regardless of what class they belong to.

Fig. 5. PCA cumulative proportion of variance by number of principal components.

The scree plot (Fig. 5) showed 80% of total variance required 56 components, 90% required 67, and 95% required 73. Meaning out of 98 features, most of them were needed to account for most of the variation in the data. There is no small group of underlying patterns driving everything. The variance is genuinely spread out, which tells us that patients cannot be described with significantly fewer numbers without losing a lot of information.

Fig. 6. PCA 2D projection of patient groupings by readmission class.

With the scree results establishing that the variance is broadly distributed, the 2D scatter (Fig. 6) shows whether the three readmission classes at least occupy different regions of the compressed space. PC1 only captures 3.43% of total variance and PC2 captures 2.64%, so the scatter is only showing 6.07% of what is actually going on in the data. Note that the scatter plot was generated from a random subsample of 5,000 patients rather than all 71,236 training records, purely for visualization purposes since plotting the full dataset would create significant overplotting. The PCA components themselves, the scree plot, the variance thresholds, and the feature loadings were all computed on the full training set. Looking at what drives each component, PC1 is almost entirely medication management: whether a patient's medications were changed, whether they were on diabetes medication at all, whether insulin or metformin was prescribed. It essentially separates patients who were being

actively managed for diabetes from those who were not. That ended up being the single biggest source of variation across 101,766 encounters, which makes sense given how central medication management is to this patient population. PC2 is driven by clinical practice complexity, specifically num_lab_procedures, number_diagnoses, time_in_hospital, and glucose serum results. That is the illness severity axis, and it lines up almost exactly with what Random Forest identified as the most predictive features.

The three readmission classes overlap almost completely across the entire plot. There is no part of the space where one class pulls away cleanly from the others. That is the most important thing taken away from this entire analysis. The three groups of patients do not look meaningfully different from each other in this feature space, at least not along the directions that carry the most information. This finding helps explain why every model hit a similar ceiling and why Class 2 was so consistently difficult to identify.

V. CUMULATIVE MODEL COMPARISON

Across all three supervised models, a consistent pattern emerged regardless of algorithm, feature set, or hyperparameter configuration. Table I summarizes the key results for each model configuration evaluated.

TABLE I. CUMULATIVE MODEL PERFORMANCE SUMMARY

Looking across all eight configurations, overall accuracy stayed between 0.49 and 0.58 regardless of model complexity, and no configuration broke through that ceiling despite using three different algorithms, two feature sets, and multiple hyperparameter values. Class 2 recall provided the most significant insights. Without class balancing, every model produced zero or near-zero Class 2 recall, confirming that unbalanced models simply learned to ignore the minority class entirely. With balancing applied, Class 2 recall improved, ranging from 0.01 in Random Forest to 0.30 in RBF SVM, but always at some cost to overall accuracy. The most notable tradeoff was between Linear SVM and RBF SVM. Linear SVM with balanced weighting achieved higher overall accuracy (0.56) but lower Class 2 recall (0.12), while RBF SVM achieved lower overall accuracy (0.49) but the highest Class 2 recall of any model (0.30). Which of those was the better model depended entirely on the clinical use case.

VI. DISCUSSION AND LIMITATIONS

The most important finding of this project was not any individual model result but the consistency of the pattern across all of them. Three different algorithms, applied with multiple configurations and balancing strategies, all plateaued at similar accuracy levels, and all struggled with Class 2 recall. The natural question I kept asking during implementation was whether a better model would have fixed this. My PCA results suggested the answer was no, at least not with this feature set. The scree plot showed that 56 components were needed to explain just 80% of the variance in the data, meaning the information in this dataset was genuinely spread across almost the entire feature space with no small group of dominant patterns driving most of the

variation. When combined with the 2D scatter plot showing near-complete overlap between all three readmission classes, the picture became clear: the three groups of patients did not look meaningfully different from each other based on the features available at discharge. That was not a modeling problem but a data structure one, and no amount of hyperparameter tuning or algorithm switching was going to change that. This also explained why reducing the feature set from 98 to 83 had almost no effect on accuracy, why Random Forest and LASSO identified the same top variables, and why adding more trees to the Random Forest did nothing. In every case, the models were working with the same underlying information, and that information was simply not strong enough to cleanly separate early readmission patients from everyone else based on structured discharge data alone.

The RBF SVM result stood out the most. It achieved the highest Class 2 recall at 0.30 while simultaneously producing the lowest overall accuracy at 0.49. This tradeoff was not a failure but suggested that what distinguished early readmission patients from others was not entirely absent, it was just non-linear and weak. The RBF kernel's ability to create curved decision boundaries allowed it to find some of that where linear models could not. A recall of 0.30 meant roughly one in three early readmission patients got flagged before discharge, which was not good enough to be clinically reliable on its own, but was meaningfully better than zero, which was what every unbalanced model produced. In a clinical environment where catching early readmission cases is the primary goal, that may actually matter more than the overall accuracy number suggested.

Originally, Strack et al. found that HbA1c measurement was associated with reduced readmission rates for patients with a primary diabetes diagnosis. This project approached the same dataset from a different angle, asking whether the full feature set could predict readmission at all. The answer suggested it largely could not, at least not with the precision needed for clinical decision support. This was not a conflict with their findings. Their study identified a meaningful statistical association between one clinical practice and one outcome. This work asked: could we predict at the individual patient level who would come back? That question turned out to require a different kind of information than what this dataset alone provided.

A. Limitations

The issue of class imbalance was never fully solved. The balancing strategy applied made a meaningful improvement over the unbalanced baseline but did not resolve the fundamental problem that Class 2 patients were structurally underrepresented. A Synthetic Minority Over-Sampling Technique (SMOTE), which generates synthetic minority class examples before training rather than reweighting during training, was not applied here due to time constraints but may have produced better Class 2 recall across all three models. That remains the most obvious next step for future iterations of this work.

Second, the RBF SVM results carried a caveat. Training on a 10,000-row subsample rather than the full 71,236 training rows meant the RBF results are not directly

comparable to the other models on equal footing. The decision boundary learned from 10,000 representative rows was likely close to what the full dataset would produce, since SVM only cares about support vectors near the margin, but it is still a constraint worth acknowledging when interpreting Table I.

Third, the dataset covered 1999 to 2008. Diabetes treatment protocols, medication options, and hospital care standards have changed substantially since then. The models trained here reflected patterns of care from over 15 years ago, which limits their direct applicability to current clinical settings. This does not invalidate the analysis as a methodological exercise, but it does mean the specific feature importances and model weights should not be interpreted as reflecting current clinical practice.

Fourth, structured electronic health record data captures what was documented at discharge, not what happened after. Patient behavior following discharge, medication adherence, access to follow-up care, and social determinants of health are among the strongest known predictors of readmission but none of them appear in this dataset. The fact that PCA could not find clean separation between the classes may be telling us less about the limits of machine learning and more about the limits of what structured discharge data can capture.

VII. CONCLUSION

Predicting early hospital readmission from structured discharge data is a harder problem than the feature richness of the dataset might initially suggest, and what this project revealed is that the difficulty is not primarily a modeling challenge but a data structure one. The features captured at discharge describe what happened during a hospitalization

well, but they say very little about the conditions a patient returns to afterward. That gap is where readmission risk may live for many patients, and it is a gap that no amount of model complexity could bridge if the underlying indicators were not there.

Coming to this work from a Psychology background, the most honest takeaway is that early readmission is a systems problem as much as a clinical one. It reflects breakdowns in factors like care transitions, external support, medication adherence, and follow-up capacity that structured electronic health records are simply not designed to capture. Future iterations of this kind of analysis would benefit most from incorporating post-discharge data, social determinants of health, and unstructured clinical notes that preserve the qualitative context that structured fields cannot. Longitudinal patient histories across multiple encounters would also add meaningful information that a single snapshot encounter could not provide. On the technical side, applying SMOTE oversampling before training rather than relying solely on class weighting is the most direct next step for improving minority class recall, and gradient boosting methods like XGBoost may also find more useful patterns in the existing features than the models tested here. Either way, the most important investment going forward remains in the data itself.

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APPENDIX A: FIGURES AND TABLES

All figures are reproduced here at full page width for clarity. Figure numbers correspond to in-text references throughout the paper.

FIGURE 1: Patient Demographics

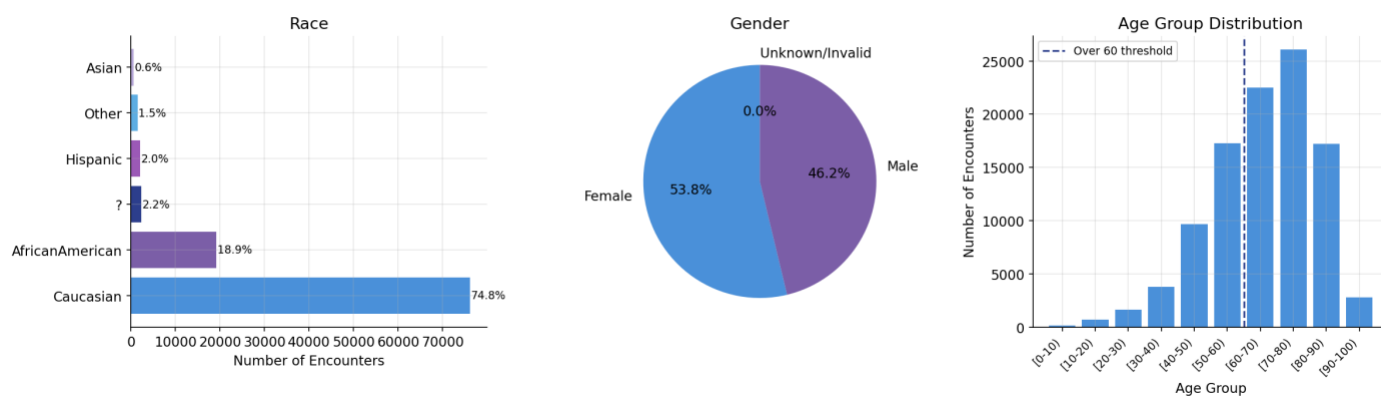


Fig. 1. Patient demographics across race, gender, and age group distribution.

FIGURE 2: Readmission Class Distribution

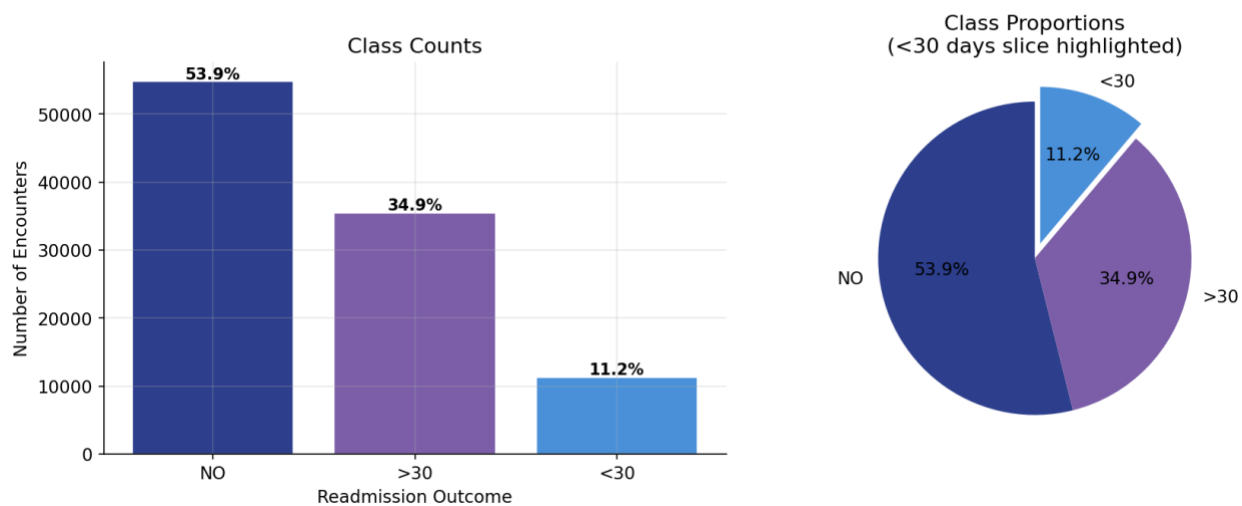
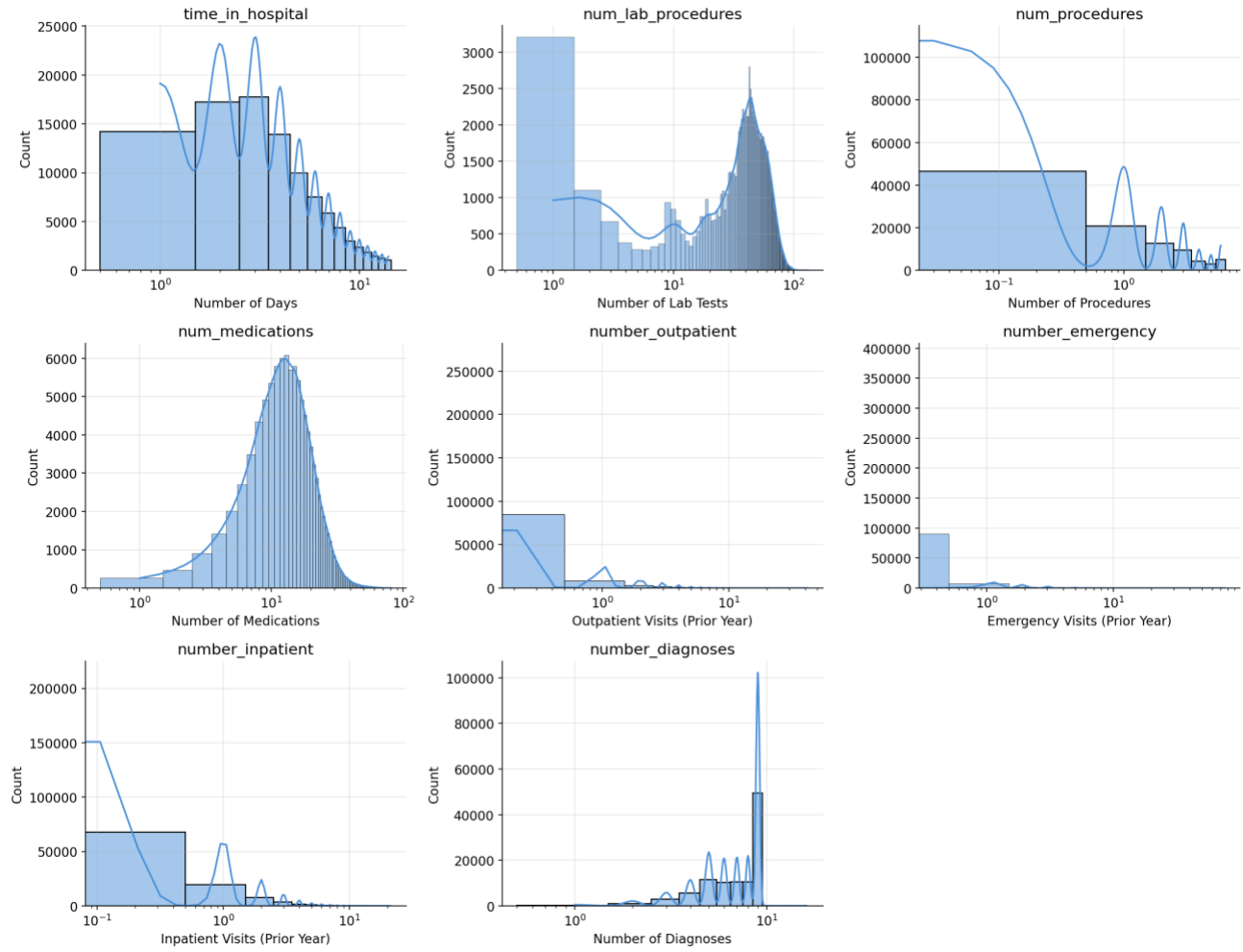


Fig. 2. Readmission class distribution showing class counts and proportions.

FIGURE 3: Distribution of Numeric Variables in Diabetic Data*Fig. 3. Distribution of numeric variables in the diabetic dataset using log-scale x-axis.*

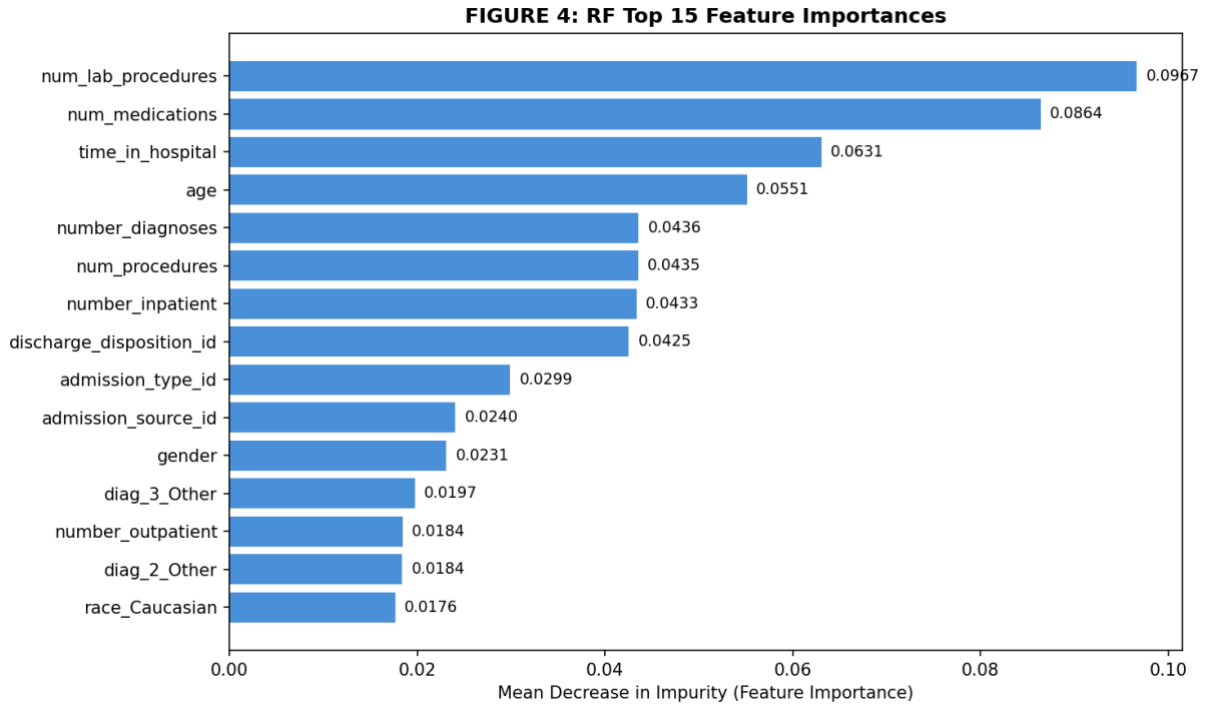


Fig. 4. Random Forest top 15 feature importances by mean decrease in impurity.

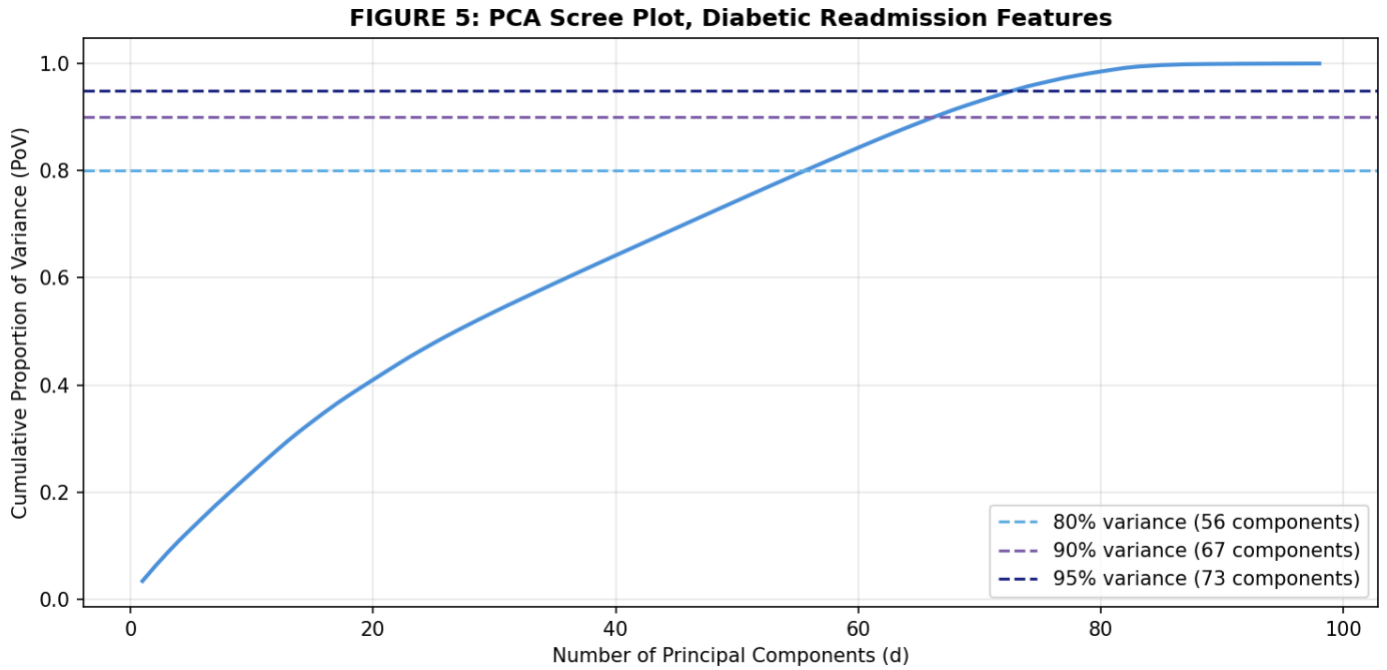


Fig. 5. PCA cumulative proportion of variance by number of principal components.

**FIGURE 6: PCA 2D Projection, Patient Groupings by Readmission Class
(5,000 patient subsample from training set)**

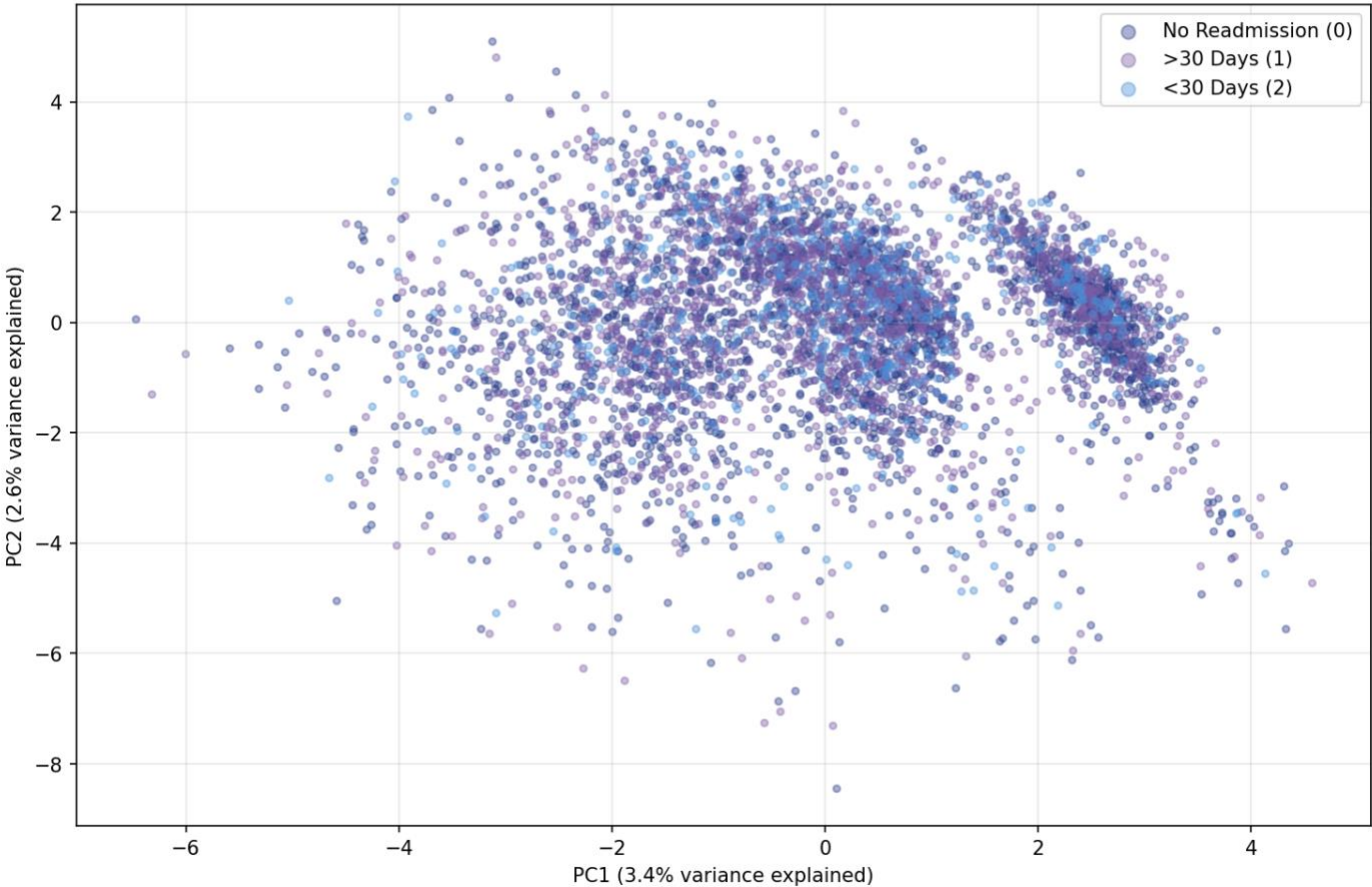


Fig. 6. PCA 2D projection of patient groupings by readmission class.

TABLE I: Cumulative Model Performance Summary

Model	Features	Accuracy	Class 0 F1	Class 1 F1	Class 2 Recall
LR Baseline (balanced)	Full (98)	0.4892	0.63	0.38	0.39
LR LASSO C=0.01	LASSO (83)	0.4884	0.63	0.38	0.38
RF Baseline (unbalanced)	Full (98)	0.5810	0.71	0.42	0.01
RF Balanced n=200	Full (98)	0.5825	0.71	0.41	0.01
RF Balanced n=200	LASSO (83)	0.5829	0.71	0.41	0.01
LinearSVC Baseline	Full (98)	0.5709	0.71	0.32	0.00
LinearSVC Balanced C=0.1	Full (98)	0.5604	0.69	0.40	0.12
RBF SVM C=0.1 (10k sub)	Full (98)	0.4898	0.62	0.40	0.30

■ Indicates highest Class 2 Recall across all configurations (RBF SVM, C=0.1)

Table 1. Cumulative Summary and Model Comparison